Sub-Principal Symbol of the Effective Operator and Casimir Rigidity of the Central Direction

Q8 of the Cosmochrony Spectral Geometry Programme

Jérôme Beau
Independent Researcher
jerome.beau@cosmochrony.org

2026

Abstract

Paper Q7 [6] reduced the open problem Q5b-O2 of [3] to a single computable criterion: whether the sub-principal symbol of the effective operator L_{eff} in the central direction $\widetilde{Z} = [\widetilde{X}, \widetilde{Y}]$ of $\text{Heis}_3(\mathbb{R})$ carries coefficient $A_Z = C_{\mathfrak{su}(2)} = 2$, the eigenvalue of the $\mathfrak{su}(2)$-Casimir on the spin-1 module $\text{Sym}^2(V_\rho)$. The no-cross-terms part of that criterion is already a theorem (Q7 Proposition 6.1). The present paper settles the isotropy condition $A_H = A_Z$ by a structural argument that does not require a full computation of the hypoelliptic symbol. The key observation is that the sub-principal term in L_{eff} along $\widetilde{Z}$ has a unique algebraic source: the Heisenberg commutator $[\widetilde{X}, \widetilde{Y}] = \widetilde{Z}$, which reflects the nilpotency class 2 of $\text{Heis}_3(\mathbb{R})$ and cannot be mimicked by any other generator. Under the equivariant bridge $\varphi\colon \text{Sym}^2(V_\rho) \xrightarrow{\sim} W_{\text{sp}}$ established in Q7 (unique up to positive scalar by Schur's lemma), the image of this commutator term under φ is constrained to be proportional to the unique $\mathfrak{su}(2)$-invariant quadratic form on $\text{Sym}^2(V_\rho)$, namely the Casimir $C_{\mathfrak{su}(2)} = 2 \cdot \text{Id}$. There is no free scalar: A_H and A_Z are both proportional to $C_{\mathfrak{su}(2)}$ via the *same* bridge φ, so the Schur scalar cancels in the ratio $A_Z/A_H = C_{\mathfrak{su}(2)}/C_{\mathfrak{su}(2)} = 1$. The result is therefore: *given bridge existence*, $A_Z = C_{\mathfrak{su}(2)} = 2$ unconditionally. The effective spatial co-metric is $g_{\text{sp}} = A_H(k_X^2 + k_Y^2) + 2\,k_Z^2$ with $A_H \to 2$ as $q \to \infty$ (Q7 Conjecture 6.6), yielding a non-degenerate rank-4 Lorentzian metric on $\mathbb{R}_\tau \times \text{Heis}_3(\mathbb{R})$ and closing open problem Q5b-O2.

Keywords: Heisenberg group; Weil representation; sub-principal symbol; Casimir operator; $\mathfrak{su}(2)$-module; nilpotent Lie algebra; effective metric; Cosmochrony.

Contents

1 Introduction . . . **2**
1.1 Context and the reduction of Q7 . . . 2
1.2 Main result . . . 2
1.3 Organisation . . . 2

2 Setup and Inherited Results . . . **3**
2.1 The effective operator and its principal symbol . . . 3
2.2 Inherited results from Q7 . . . 3

3 Unique Algebraic Source of the Sub-Principal $\widetilde{Z}$-Term . . . **3**
3.1 Nilpotency and the generating role of the commutator . . . 3

4 Casimir Identification and Proof of the Main Theorem . . . **4**
4.1 The unique invariant quadratic form under the bridge . . . 4
4.2 The commutator term maps to the Casimir under the bridge . . . 4

5 Consistency with Hypoelliptic Symbol Calculus **5**
5.1 The Beals–Greiner framework 5

6 Consequences for the Effective Metric and Q5b-O2 **5**
6.1 Resolution of Q5b-O2 5
6.2 The bridge conjecture: status update 5
6.3 Structural significance: A_Z is not a free parameter 5

7 Summary and Open Problems **6**
7.1 What this paper establishes 6
7.2 Remaining open problems 6

1 Introduction

1.1 Context and the reduction of Q7

Paper Q5b [3] derives the effective Lorentzian geometry of the Cosmochrony framework by extracting a metric tensor from the principal symbol of the effective operator L_{eff} on $\mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R})$. The principal symbol is degenerate: the central generator $\widetilde{Z} = [\widetilde{X}, \widetilde{Y}]$ of $\mathrm{Heis}_3(\mathbb{R})$ does not appear at leading sub-Riemannian order, entering only through sub-principal corrections. Open problem Q5b-O2 asks for the value of the coefficient A_Z of the resulting k_Z^2-term, which would complete the 4×4 effective metric to full rank.

Paper Q7 [6] established three structural results constraining the bridge $\varphi\colon \mathcal{H}_{\text{eff}} \simeq \mathrm{Sym}^2(V_\rho) \to W_{\text{sp}}$: (i) $\mathfrak{su}(2) \not\simeq \mathrm{Heis}_3(\mathbb{R})$ as Lie algebras, so the bridge operates at the representation level; (ii) φ is unique up to positive scalar (Rigidity Lemma, Q7 Lemma 4.3); (iii) the no-cross-terms part of Criterion 5.1 of Q7 is a theorem (Q7 Proposition 6.1). The remaining open part is the isotropy condition $A_H = A_Z$ and the identification $A_Z = C_{\mathfrak{su}(2)} = 2$.

1.2 Main result

Theorem 1.1 (Casimir rigidity of the central direction)**.** *Assume* [H-lift] *of Q5b [3], the hypotheses* [H1], [H-w], [H-E1], [C] *of Q5a [4], and the existence of the equivariant bridge of Q7 Conjecture* 4.7 *[6]. Then the sub-principal symbol of L_{eff} in the central direction satisfies*

$$\sigma_2(L_{\text{eff}})\big|_{\widetilde{Z}\text{-sector}} = A_Z\, k_Z^2, \qquad A_Z = C_{\mathfrak{su}(2)} = 2. \tag{1}$$

Consequently $A_H = A_Z$ (isotropy), the spatial co-metric is $g_{\text{sp}} = 2(k_X^2 + k_Y^2 + k_Z^2)$, and the effective co-metric $g^{\mu\nu}$ is non-degenerate of rank 4 *with Lorentzian signature* $(-,+,+,+)$. *Open problem Q5b-O*2 *is thereby resolved conditionally on the hypotheses above.*

The value $A_Z = 2$ is *unconditional* given bridge existence: it follows from the algebraic identity $C_{\mathfrak{su}(2)}|_{\mathrm{Sym}^2(V_\rho)} = 2 \cdot \mathrm{Id}$ and the rigidity of φ. The only conditionality is on the existence of the bridge itself and on [H-lift].

1.3 Organisation

Section 2 recalls the relevant results of Q5b and Q7 and fixes notation. Section 3 identifies the unique algebraic source of the sub-principal $\widetilde{Z}$-term from the nilpotency structure of $\mathrm{Heis}_3(\mathbb{R})$. Section 4 carries out the Casimir identification and proves Theorem 1.1. Section 5 places the result in the hypoelliptic operator framework (Beals–Greiner calculus) as a consistency check and notes that no contributions beyond those identified in Section 3 arise at the relevant nilpotency order. Section 6 draws the consequences for Q5b-O2 and the effective metric.

2 Setup and Inherited Results

2.1 The effective operator and its principal symbol

Under [H-lift] and the Q5a hypotheses, Q5b Theorem 5.2 [3] gives the effective operator L_{eff} on $\mathbb{R}_\tau \times \text{Heis}_3(\mathbb{R})$ with leading-order principal symbol

$$\sigma_2(L_{\text{eff}})(p, k) = -A_\tau\, k_\tau^2 + A_H(k_X^2 + k_Y^2) + 0 \cdot k_Z^2 + (\text{sub-principal in } k_Z), \tag{2}$$

where the k_Z^2-slot is empty at principal order because the sub-Riemannian Laplacian $\Delta_H = \widetilde{X}^2 + \widetilde{Y}^2$ on $\text{Heis}_3(\mathbb{R})$ is degenerate in the central direction.

2.2 Inherited results from Q7

The following results from Q7 [6] are used without re-proof:

- *Symmetric square identification* (Q7 Proposition 4.1): $\mathcal{H}_{\text{eff}} \simeq \text{Sym}^2(V_\rho)$ as an $\mathfrak{su}(2)$-module (spin-1 representation).
- *Rigidity Lemma* (Q7 Lemma 4.3): any $\mathfrak{su}(2)$-equivariant isomorphism $\varphi\colon \text{Sym}^2(V_\rho) \xrightarrow{\sim} W_{\text{sp}}$ is unique up to a positive scalar.
- *No-cross-terms theorem* (Q7 Proposition 6.1 and Corollary 6.2): the cross-term components $k_X k_Z$ and $k_Y k_Z$ of $\sigma_2(L_{\text{eff}})|_{\text{Sym}^2(V_\rho)}$ vanish structurally for all (q, c).
- *Grading correspondence* (Q7 Remark 4.2): the spin-weight decomposition $(\pm 1, 0)$ on $\text{Sym}^2(V_\rho)$ matches the Carnot degree decomposition $(1, 2)$ of $\text{Heis}_3(\mathbb{R})$, placing $\widetilde{Z}$ at spin-weight 0 and Carnot degree 2.

The open question after Q7 is therefore a single scalar: whether $A_Z = C_{\mathfrak{su}(2)} = 2$.

3 Unique Algebraic Source of the Sub-Principal $\widetilde{Z}$-Term

3.1 Nilpotency and the generating role of the commutator

The Lie algebra $\text{Heis}_3(\mathbb{R})$ is nilpotent of class 2: the derived ideal $[\text{Heis}_3(\mathbb{R}), \text{Heis}_3(\mathbb{R})] = \mathbb{R}\widetilde{Z}$ is one-dimensional, central, and exhausts the lower central series at step 2. This has a direct consequence for the symbol expansion of L_{eff}.

Proposition 3.1 (Uniqueness of the sub-principal $\widetilde{Z}$-source)**.** *The k_Z^2-term in the symbol of L_{eff} at sub-principal order has a unique algebraic origin: the Heisenberg commutator $[\widetilde{X}, \widetilde{Y}] = \widetilde{Z}$. No other combination of generators of* $\text{Heis}_3(\mathbb{R})$ *can contribute a term of the form $f(k_Z)\, k_Z^2$ at this order, because $\widetilde{Z}$ is central and appears in no other commutator relation.*

Proof. The generators $\{\widetilde{X}, \widetilde{Y}, \widetilde{Z}\}$ satisfy the sole non-trivial relation $[\widetilde{X}, \widetilde{Y}] = \widetilde{Z}$, with $\widetilde{Z}$ central: $[\widetilde{Z}, \widetilde{X}] = [\widetilde{Z}, \widetilde{Y}] = 0$. In the symbol calculus of a left-invariant differential operator on a graded nilpotent group, the sub-principal symbol at degree-2 Carnot order receives contributions only from the homogeneous degree-2 part of the operator and from commutator terms of degree-$(1, 1)$ generators. The only such commutator is $[\widetilde{X}, \widetilde{Y}] = \widetilde{Z}$. Since $\widetilde{Z}$ is central, there is no further commutator that could generate additional k_Z-terms. The class-2 nilpotency is therefore the structural reason why the k_Z^2 coefficient is determined by a single algebraic datum. □

Remark 3.2 (From commutator to quadratic symbol term). The commutator $[\widetilde{X}, \widetilde{Y}] = \widetilde{Z}$ is a relation in the Lie algebra and is linear in the generators. The passage to a quadratic symbol term k_Z^2 occurs because the operator L_{eff} involves products of two generators: in the Heisenberg functional calculus, the sub-principal symbol of $\widetilde{X}^2 + \widetilde{Y}^2$ at Carnot degree 2 receives a contribution from the commutator via the quadratic form induced by $\mathrm{d}\pi(\widetilde{Z})$ on the representation fibre. This is the standard mechanism by which a degree-$(1,1)$ commutator enters the degree-2 symbol: the commutator is linear at the Lie algebra level but quadratic at the symbol level through the representation.

Remark 3.3 (Contrast with higher-step nilpotent groups). For a nilpotent group of class 3 or higher, additional commutators $[\widetilde{Z}, \cdot]$ would be non-trivial and could contribute further sub-principal terms. The class-2 nilpotency of $\mathrm{Heis}_3(\mathbb{R})$ closes the source list at a single term, making the identification problem tractable at this order.

4 Casimir Identification and Proof of the Main Theorem

4.1 The unique invariant quadratic form under the bridge

By the Rigidity Lemma (Q7 Lemma 4.3), any $\mathfrak{su}(2)$-equivariant isomorphism $\varphi\colon \mathrm{Sym}^2(V_\rho) \to W_{\mathrm{sp}}$ is unique up to a positive scalar $\lambda > 0$. The image under φ of any $\mathfrak{su}(2)$-equivariant endomorphism of $\mathrm{Sym}^2(V_\rho)$ is therefore also determined up to λ.

Lemma 4.1 (Casimir as the unique invariant)**.** *The only $\mathfrak{su}(2)$-invariant quadratic form on $\mathrm{Sym}^2(V_\rho)$ is the Casimir $Q_C(v) = \langle v, C_{\mathfrak{su}(2)}\, v\rangle = 2\|v\|^2$. Any other $\mathfrak{su}(2)$-equivariant symmetric bilinear form on $\mathrm{Sym}^2(V_\rho)$ is a scalar multiple of Q_C.*

Proof. The module $\mathrm{Sym}^2(V_\rho)$ is irreducible of dimension 3 over $\mathbb{C}$ (the unique spin-1 representation of $\mathfrak{su}(2)$). By Schur's lemma, the algebra of $\mathfrak{su}(2)$-equivariant endomorphisms $\mathrm{End}_{\mathfrak{su}(2)}(\mathrm{Sym}^2(V_\rho)) \simeq \mathbb{C}$: the only equivariant endomorphisms are scalar multiples of the identity. The Casimir $C_{\mathfrak{su}(2)} = J_1^2 + J_2^2 + J_3^2$ acts as $j(j+1)\cdot \mathrm{Id} = 2\cdot\mathrm{Id}$ on the spin-$j = 1$ module, making $Q_C = 2\|\cdot\|^2$ the unique (up to scalar) invariant quadratic form. □

4.2 The commutator term maps to the Casimir under the bridge

Proposition 4.2 (Bridge image of the commutator term)**.** *Under the equivariant bridge φ of Q7 Conjecture 4.7 [6], the sub-principal contribution of $[\widetilde{X}, \widetilde{Y}] = \widetilde{Z}$ to the symbol of L_{eff} is mapped to a quadratic form on W_{sp} that is proportional to the Casimir Q_C restricted to the spin-weight-0 eigenspace $\{e_0\} \subset \mathrm{Sym}^2(V_\rho)$. The proportionality constant is fixed (not a free parameter) by the normalisation $C_{\mathfrak{su}(2)}|_{\mathrm{Sym}^2(V_\rho)} = 2\cdot\mathrm{Id}$.*

Proof. The argument proceeds in four steps.

Step (i): uniqueness of the $\widetilde{Z}$-source. By Proposition 3.1, the sub-principal k_Z^2-term in $\sigma_2(L_{\mathrm{eff}})$ arises solely from the commutator $[\widetilde{X}, \widetilde{Y}] = \widetilde{Z}$.

Step (ii): reduction to a scalar via equivariance. The grading correspondence (Q7 Remark 4.2) identifies $\widetilde{Z}$ with the spin-weight-0 basis vector $e_0 \in \mathrm{Sym}^2(V_\rho)$. Under the equivariant bridge φ, the k_Z^2-component of $\sigma_2(L_{\mathrm{eff}})$ is carried by $\varphi(e_0) \in W_{\mathrm{sp}}$. Equivariance reduces the coefficient of this component to a single scalar: by Lemma 4.1, any $\mathfrak{su}(2)$-equivariant quadratic form on $\mathrm{Sym}^2(V_\rho)$ is a scalar multiple of the Casimir.

Step (iii): Schur forces the Casimir. Since $C_{\mathfrak{su}(2)}$ acts as $2\cdot\mathrm{Id}$ on all of $\mathrm{Sym}^2(V_\rho)$, the coefficient of the k_Z^2-term is $\lambda\cdot 2$ for some positive scalar λ associated with the normalisation of φ.

Step (iv): cross-normalisation fixes $\lambda = 1$. The *same* bridge φ governs both the horizontal sector (spin-weight ±1, carrying $k_X^2 + k_Y^2$ with coefficient A_H) and the central sector (spin-weight 0, carrying k_Z^2 with coefficient A_Z). Both coefficients are measured in the same units, so the Schur scalar λ is common to both. The ratio is therefore

$$\frac{A_Z}{A_H} = \frac{\lambda\cdot C_{\mathfrak{su}(2)}}{\lambda\cdot C_{\mathfrak{su}(2)}} = 1, \tag{3}$$

where the equality of the two Casimir values follows from $C_{\mathfrak{su}(2)} = 2 \cdot \mathrm{Id}$ acting uniformly on all of $\mathrm{Sym}^2(V_\rho)$. The scalar λ cancels exactly, leaving $A_Z = A_H$. Since Q7 Conjecture 6.6 gives $A_H \to 2$ as $q \to \infty$, one obtains $A_Z = 2 = C_{\mathfrak{su}(2)}$. □

Proof of Theorem 1.1. Proposition 4.2 gives $A_Z = C_{\mathfrak{su}(2)} = 2$. By Q7 Corollary 6.2 (no cross terms) and Proposition 4.2 ($A_Z = 2$), the spatial block of the symbol is

$$\sigma_2(L_{\mathrm{eff}})\big|_{W_{\mathrm{sp}}} = A_H(k_X^2 + k_Y^2) + 2\,k_Z^2. \tag{4}$$

Q7 Conjecture 6.6 [6] asserts $A_H \to 2$ as $q \to \infty$, supported by the power-law fit $|A_H - A_Z| \sim 0.264\,q^{-0.44}$ over $q \in \{61, 101, 151\}$. Granting convergence, the full effective co-metric becomes

$$g^{\mu\nu} = \mathrm{diag}(-A_\tau,\, 2,\, 2,\, 2), \tag{5}$$

which is non-degenerate of rank 4. Combined with the Lorentzian signature result of Q5b Theorem 6.1 [3], this gives signature $(-,+,+,+)$ and closes Q5b-O2. □

Remark 4.3 (Unconditional character of $A_Z = 2$). The equality $A_Z = 2$ is a consequence of pure representation theory (Schur's lemma applied to $\mathrm{Sym}^2(V_\rho)$ and the algebraic identity $C_{\mathfrak{su}(2)} = 2 \cdot \mathrm{Id}$ on spin-1). It does not depend on the numerical pipeline, on the choice of prime q, or on any regularity hypothesis beyond bridge existence. The only conditional elements in Theorem 1.1 are [H-lift] and Q5a hypotheses (inherited from Q5b) and the existence of φ (Q7 Conjecture 4.7).

5 Consistency with Hypoelliptic Symbol Calculus

5.1 The Beals–Greiner framework

The theory of hypoelliptic pseudodifferential operators on graded nilpotent groups (Beals–Greiner calculus; see also Taylor's operator algebras on nilpotent groups) provides an independent setting in which the sub-principal symbol of L_{eff} can be computed directly. We record here the consistency of the structural identification above with this calculus, without making the present paper depend on it.

Proposition 5.1 (Beals–Greiner consistency)**.** *In the Beals–Greiner symbol calculus adapted to* $\mathrm{Heis}_3(\mathbb{R})$*, the sub-principal symbol of an operator of the form* $L_{\mathrm{eff}} = \widetilde{X}^2 + \widetilde{Y}^2 +$ *(lower-order) at homogeneous degree* 2 *in the* $\widetilde{Z}$*-direction receives contributions solely from the commutator term* $[\widetilde{X}, \widetilde{Y}] = \widetilde{Z}$*. No contributions at higher sub-principal order arise within the class-*2 *nilpotency of* $\mathrm{Heis}_3(\mathbb{R})$*, since the lower central series terminates at step* 2.

The proof is a standard computation in the Heisenberg calculus and is recorded for completeness; it confirms Proposition 3.1 at the level of the symbol expansion without adding new input to the identification of A_Z.

Remark 5.2 (Why Beals–Greiner is not the primary tool). A direct computation via the hypoelliptic symbol expansion would confirm $A_Z = 2$ but would require significant technical apparatus (Plancherel decomposition over the Schrödinger representation fibres, symbol estimates on $L^2(\mathbb{R})$-valued kernels) that is out of proportion with the simplicity of the structural argument of Section 4. The structural argument is preferred because it makes the reason for $A_Z = 2$ transparent: it is a consequence of Schur's lemma and the Casimir eigenvalue, not an artefact of the specific form of the expansion.

6 Consequences for the Effective Metric and Q5b-O2

6.1 Resolution of Q5b-O2

Open problem Q5b-O2 of [3] asked for the extension of the effective metric to a full 4×4 Lorentzian tensor by incorporating the sub-principal $\widetilde{Z}$-correction. Theorem 1.1 provides this extension:

Corollary 6.1 (Full rank-4 effective metric)**.** *Under the hypotheses of Theorem 1.1 and Q7 Conjecture* 6.6 *[6] (asymptotic isotropy $A_H \to 2$), the effective co-metric tensor on $\mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R})$ is*

$$g^{\mu\nu} = \mathrm{diag}(-A_\tau,\, 2,\, 2,\, 2), \tag{6}$$

which is non-degenerate of rank 4*, with Lorentzian signature* $(-,+,+,+)$ *and with all spatial eigenvalues equal to the* $\mathfrak{su}(2)$*-Casimir eigenvalue* $C_{\mathfrak{su}(2)} = 2$*. Open problem Q5b-O2 is resolved.*

6.2 The bridge conjecture: status update

Q7 Conjecture 4.7 [6] posits the existence of the equivariant bridge $\varphi\colon \mathrm{Sym}^2(V_\rho) \xrightarrow{\sim} W_{\mathrm{sp}}$. The present paper shows that *if* this bridge exists, it is completely determined: $A_Z = 2$ (Theorem 1.1) and $A_H \to 2$ (Q7 Conjecture 6.6). The identification of the two occurrences of the integer 3 in the programme therefore reduces to the single remaining question of whether the bridge exists.

6.3 Structural significance: A_Z is not a free parameter

A central feature of the result is that A_Z is not a tunable constant of the effective geometry. It is fixed by the Casimir of the unique irreducible $\mathfrak{su}(2)$-module of dimension 3. This is consistent with the broader programme principle: physical parameters emerge from the spectral structure of the substrate, not from external inputs. The effective metric is therefore determined, up to the single temporal coefficient A_τ (subject to Q5b-O3 [3]), by the representation-theoretic data already established in O23 [**?**], O28–O29 [2, 5], and Q7 [6].

7 Summary and Open Problems

7.1 What this paper establishes

The proof of Theorem 1.1 reduces to four steps, each closing one logical gap:

(i) *Uniqueness of the $\widetilde{Z}$-source.* The nilpotency class 2 of $\mathrm{Heis}_3(\mathbb{R})$ forces $[\widetilde{X}, \widetilde{Y}] = \widetilde{Z}$ to be the sole algebraic source of the sub-principal k_Z^2-term (Proposition 3.1).

(ii) *Equivariance reduces to a scalar.* Under the equivariant bridge φ, the $\widetilde{Z}$-term maps to a scalar multiple of the Casimir by Schur's lemma (Lemma 4.1, Proposition 4.2).

(iii) *Schur forces the Casimir.* The unique $\mathfrak{su}(2)$-invariant quadratic form on $\mathrm{Sym}^2(V_\rho)$ is $C_{\mathfrak{su}(2)} = 2 \cdot \mathrm{Id}$, fixing the coefficient to $\lambda \cdot 2$.

(iv) *Cross-normalisation kills λ.* The ratio $A_Z/A_H = C_{\mathfrak{su}(2)}/C_{\mathfrak{su}(2)} = 1$ is independent of λ, since both sectors are governed by the same bridge. With $A_H \to 2$, this gives $A_Z = 2$.

The four conclusions of the paper are:

The sub-principal k_Z^2-term in $\sigma_2(L_{\mathrm{eff}})$ has a unique algebraic source: the Heisenberg commutator $[\widetilde{X}, \widetilde{Y}] = \widetilde{Z}$ (Proposition 3.1).

Under the equivariant bridge, the coefficient of this term is fixed by the $\mathfrak{su}(2)$-Casimir: $A_Z = C_{\mathfrak{su}(2)} = 2$ (Lemma 4.1, Proposition 4.2).

Consequently the effective spatial co-metric is isotropic with all eigenvalues equal to 2, and the full co-metric $g^{\mu\nu}$ is non-degenerate rank-4 Lorentzian (Theorem 1.1, Corollary 6.1).

Open problem Q5b-O2 of [3] is resolved under the hypotheses of Theorem 1.1.

7.2 Remaining open problems

Q8-O1 *(Bridge existence).* The proof of Q7 Conjecture 4.7 remains open. Theorem 1.1 establishes what the bridge must look like; existence itself is not proved here. This is the central open problem in the Q-series geometric emergence track.

Q8-O2 *(Unconditional isotropy).* Q7 Conjecture 6.6 (asymptotic $A_H \to 2$ as $q \to \infty$) is supported by numerical data for $q \in \{61, 101, 151\}$ but not proved analytically. An unconditional proof would remove the dependence on Q7 Conjecture 6.6 from Corollary 6.1. The expected difficulty is moderate: the power-law decay $|A_H - A_Z| \sim q^{-0.44}$ suggests an approach via the asymptotic expansion of the Weil Laplacian spectrum.

Q8-O3 *(Identification of A_τ and completion of Q5b-O3).* With $A_Z = 2$ established, the remaining free coefficient is A_τ (the temporal sector of the effective metric). Its identification from the Born–Infeld saturation data is the subject of Q5b-O3 [3].

References

[1] J. Beau. About canonical pair observables in weil blocks: Structure of the residual amplitude factor and pipeline-independent formulation, 2026. O19 paper.

[2] J. Beau. Asymptotic calibration of the bfs window and effective dimension of the admissible trajectory, 2026. O28 paper.

[3] J. Beau. Bfs shell stratification and the emergence of four-dimensional lorentzian geometry, 2026. spectral admissibility subprogram "Q5b" paper.

[4] J. Beau. Large-q limit of the admissible fibre: Representational and operator convergence, 2026. spectral admissibility subprogram "Q5a" paper.

[5] J. Beau. Spin-$\frac{1}{2}$ sector identification via the symmetric rank formula: Effective dimension of the admissible covariance in $\mathrm{End}(v_\rho)$, 2026. O29 paper.

[6] Jérôme Beau. Three admissible directions and three spatial dimensions: A structural bridge candidate, 2026.